

UMC 0.11um Logic and Mixed-Mode AI Advanced Enhancement MIM Capacitor SPICE Model Document - Monte Carlo -

**Version 1.0 Phase 1
2016/06/15**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.2_p1	2010/07/13	Ian Chen	1. This Monte Carlo model including both process and mis-matching characteristics is extracted from l110ae_mimcap_v021.mdl l110ae_mimcap_v021.lib
0.3_p1	2010/12/09	Hung Wei Liao	1. This Monte Carlo model including both process and mis-matching characteristics is extracted from l110ae_mimcap_v031.mdl l110ae_mimcap_v031.lib 2. Add 1.5fF, 2.0fF MIM Monte Carlo Model
0.4_p1	2012/09/28	Ji Wei Hsu	1. Modify 2.0fF MIM Monte Carlo Model 2. This Monte Carlo model including both process and mis-matching characteristics is extracted from l110ae_mimcap_v041.mdl l110ae_mimcap_v041.lib
0.5_P1	2013/08/13	Ting Huei Wang	1. Change 20f MIMCAP Model minimum size from $4.8*4.8(\mu m^2)$ to $3.4*3.4(\mu m^2)$ 2. This Monte Carlo model including both process and mis-matching characteristics is extracted from l110ae_mimcap_v051.mdl l110ae_mimcap_v051.lib
0.6_p1	2015/4/22	Zung Jie Han	1. Change 20f MIMCAP Model "cf" parameter from $4.96e-11$ to $4.95e-11$ 2. This Monte Carlo model including both process and mis-matching characteristics is extracted from l110ae_mimcap_v061.mdl l110ae_mimcap_v061.lib
1.0_p1	2016/05/03	Tzu Kang Hsiung	1. Modify 15f and 10f MIMCAP Model based on new silicon data . 2. Add trend chart of mapping data into the baseband document . 3. Revise the verification figure (with VC & with TC) of 2.0fF MIMCAP in the baseband document . 4. This Monte Carlo model including both process and mis-matching characteristics is extracted from

			l110ae_mimcap_v101.mdl l110ae_mimcap_v101.lib
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1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

UMC 0.11 um Logic and Mixed-Mode AI Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

The available capacitors are:

- a. 1.0fF/um² MIM capacitors: mimcaps_10f_mml110e
- b. 1.5fF/um² MIM capacitors: mimcaps_15f_mml110e
- c. 2.0fF/um² MIM capacitors: mimcaps_20f_mml110e

1.3 Model Usage

1.3.1 Simulators

The compact model is provided in various formats. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2013.3sp2
Cadence SPECTRE version 13.1 (13.1.0.066)
Mentor Graphics ELDO version 7.9_1.1 (2013.1)

1.3.2 The definitions of sub-circuit model parameters

After invoking Monte-Carlo model parameters, the compact capacitor models evolve to sub-circuit models. The input parameters for sub-circuit model are the same as compact model, except adding two more input parameters (mf & mis_flag) for sub-circuit model.

The following is the parameter list:

m: number of devices
mf: the same as "m"
mis_flag: the flag parameter to turn on/off mismatching parameters for individual device. 1 is turn-on and 0 is turn-off. Default is 1.
w: capacitor width
l: capacitor length

The following examples show how to use these two additional parameters:

For parallel capacitor devices:

x1 (n1 n2) mimcaps_15f_mral w=W1 l=L1 mis_flag=1 M='(3)*(1)'
MF='(3)*(1)'

The above example shows 3 capacitors connected in parallel, each capacitor has width W1 and length L1. The mismatching parameters are turned-on for these 3 capacitors.(see the Fig 1-1)

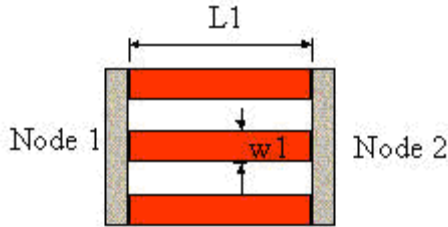


Figure 1-1 Parallel capacitor example

1.3.3 The combination of statistical model & corner model

The Monte-Carlo model and corner model (3 corners) are combined into a single model library. User can choose either the Monte-Carlo model or a corner case by selecting different model sections for simulation. The following sections are included:

mc: Monte-Carlo model
tt: Typical capacitor value
ss : Maximum capacitor value
ff: Minimum capacitor value

1.3.4 Mismatching parameters in corner model

The mismatching parameters are included in the corner models. Therefore, user can run Monte-Carlo mismatching simulation over a corner model. For example, user can evaluate the mismatching performance at the ss case.

1.3.5 Sigma number for Monte-Carlo model

User can define standard deviation at sigma level for Monte-Carlo simulation. If a larger sigma number is used, it means the model parameters will have greater variation. Sigma=3 is recommended to be used.

1.3.6 Model usage

For Synopsys HSPICE

-Include library file and select section:

Add the following statement in the input file to the simulator.

Example 1: .lib "l110ae_mimcap_v101_mc_corner.lib" mc

Example 2: .lib "l110ae_mimcap_v101_mc_corner.lib" tt

-Define standard deviation at the sigma level and pick up a value (1~3) of sigma to run MC simulations;

Example: .param sigma=3

Add the following statement in the input file to the simulator for 90% dimension shrinkage.

.Option scale=0.9

Add the following statement in the input file to the simulator for corner cases.

.lib "/ModelFileName.lib" CaseName

, where CaseName could be MC, tt , ss , ff .

-Define device condition by 'Xxx' statement;

Example : For Number device

X1 n1 n2 mimcaps_15f_mml110e w=1.00E-05 l=1.00E-05 MF=2

Example: For mismatch flag

X1 n1 n2 mimcaps_15f_mml110e w=1.00E-05 l=1.00E-05 MIS_FLAG=1
MF=2

-Define the control of Monte-Carlo simulation;

Indicator:

PROCESS: 1 to turn on the MC process simulation

0 to turn off the MC process simulation

MISMATCH: 1 to turn on the MC mismatch simulation

0 to turn off the MC mismatch simulation

Example 1:

.PARAM PROCESS=1 MISMATCH=1 * turn on MC process & mismatch simulations at the same time.

Example 2:

.PARAM PROCESS=1 MISMATCH=0 * just turn on the MC process simulation only.

Example 3:

.PARAM PROCESS=0 MISMATCH=1 * just turn on the MC mismatch simulation only.

-Define the number of Monte Carlo iterations to run MC simulations;

Example: .DC Vgg 0 1.0 0.01 Sweep Monte=200

For Cadence Spectre

-Include library file and select section:.

Example 1: include "l110ae_mimcap_v101_mc_corner.lib.scs" section=mc

Example 2: include "l110ae_mimcap_v101_mc_corner.lib.scs" section=tt

-Define standard deviation at the sigma level and pick up a value (1~3) of sigma to run MC simulations;

Example: parameters sigma=3

- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

SetOption1 options scale=0.9

-Add the following statement in the input file to the simulator for corner cases.

include "./ModelFileName.lib.scs" section=CaseName

, where CaseName could be MC, tt , ss , ff.

-Define device condition by 'Xxx' statement;

Example : For Number device

X1 (n1 n2) mimcaps_15f_mml110e w=1.0E-05 l=1.0E-05 mf=2

Example: For mismatch flag

X1 (n1 n2) mimcaps_15f_mml110e w=1.0E-05 l=1.0E-05 mis_flag=1 mf=2

-Define the number of Monte Carlo iterations and control MC simulations;

Example 1: run mismatch simulation only

mc1 montecarlo numruns=10 seed=1 variations=mismatch donominal=yes
savefamilyplots=yes { analysis DC1 dc param=Vgg start=0 stop=1.0 step=0.01 }

Example 2: run process simulation only

mc1 montecarlo numruns=10 seed=1 variations=process donominal=yes
savefamilyplots=yes { analysisDC1 dc param=Vgg start=0 stop=1.0 step=0.01 }

Example 3: run process & mismatch simulation together

mc1 montecarlo numruns=10 seed=1 variations=all donominal=yes
savefamilyplots=yes { analysisDC1 dc param=Vgg start=0 stop=1.0 step=0.01 }

For Mentor ELDO

-Include library file and select section:.

Example 1: .lib "l110ae_mimcap_v101_mc_corner.lib.eldo" mc

Example 2: .lib "l110ae_mimcap_v101_mc_corner.lib.eldo" tt

-Define standard deviation at the sigma level and pick up a value (1~3) of sigma to run MC simulations;

Example: .param sigma=3

- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

.Option scale=0.9

-Add the following statement in the input file to the simulator for corner cases.

.lib "/ModelFileName.lib.eldo" CaseName

, where CaseName could be MC, tt , ss , ff.

-Define device condition by 'Xxx' statement;

Example : For Number device

X1 n1 n2 mimcaps_15f_mml110e w=1.00E-05 l=1.00E-05 mf=2

Example: For mismatch flag

X1 n1 n2 mimcaps_15f_mml110e w=1.00E-05 l=1.00E-05 MIS_FLAG=1

mf=2

-Define the control of Monte-Carlo simulation;

Indicator:

PROCESS: 1 to turn on the MC process simulation

0 to turn off the MC process simulation

MISMATCH: 1 to turn on the MC mismatch simulation

0 to turn off the MC mismatch simulation

Example 1:

.PARAM PROCESS=1 MISMATCH=1 * turn on MC process & mismarch simulations at the same time.

Example 2:

.PARAM PROCESS=1 MISMATCH=0 * just turn on the MC process simulation only.

Example 3:

.PARAM PROCESS=0 MISMATCH=1 * just turn on the MC mismatch simulation only.

-Define the number of Monte Carlo iterations to run MC simulations;

Example: .DC Vgg 0 1.0 0.01 Sweep Monte=200

ModelFileName: l110ae_mimcap_v101_mc

DeviceName for 1.0fF/um2 MIM capacitor: mimcaps_10f_mml110e.

for 1.5fF/um2 MIM capacitor: mimcaps_15f_mml110e

for 2.0fF/um2 MIM capacitor: mimcaps_20f_mml110e.

As stated above that one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry Range(before 90% shrink):

w: 3.4um ~ 49.2um

l: 3.4um ~49.2um

Temperature range:

-50 °C ~ +125 °C

2 Monte Carlo DC Simulation Results for Process Variation

Monte Carlo iterations is 1000 times to be performed in following simulation condition. Reasonable Monte Carlo iterative times are 30. The statistical significance of 30 iterations is quite high. If the circuit operates correctly for all 30 iterations, there is 99% probability that over 80% of all possible component values operate correctly. The relative error of a quantity determined through Monte Carlo analysis is proportional to (iteration times)^{-1/2}.

2.1 Monte Carlo Simulation Result Plots

The following figures show the Monte Carlo simulation result of Capacitor at 25 °C

Capacitor Type (W/L=49.2/49.2um)	TT	MIN	MAX
1.0fF/um ² MIM Capacitor (F)	1.9750E-12	1.6082E-12	2.3225E-12
1.5fF/um ² MIM Capacitor (F)	2.9470E-12	2.4937E-12	3.4029E-12
2.0fF/um ² MIM Capacitor (F)	3.9349E-12	3.3519E-12	4.8252E-12

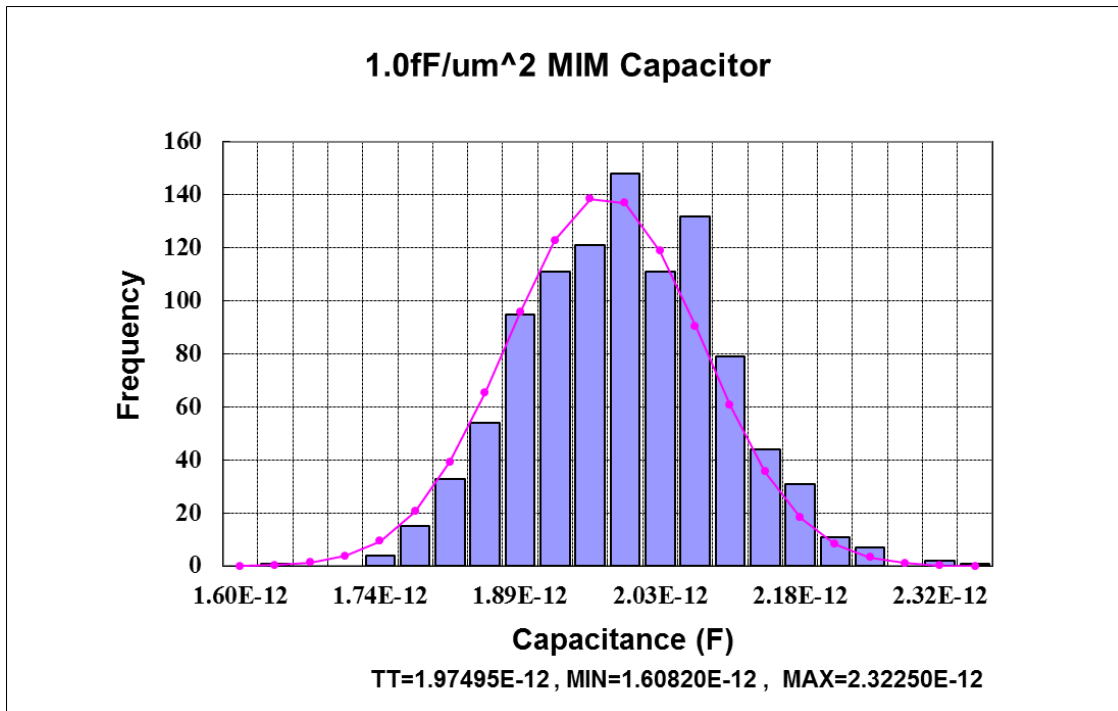


Figure 2-1 Monte Carlo plots of W/L=49.2um/49.2um (before shrink) 1.0fF/um² MIM Capacitor

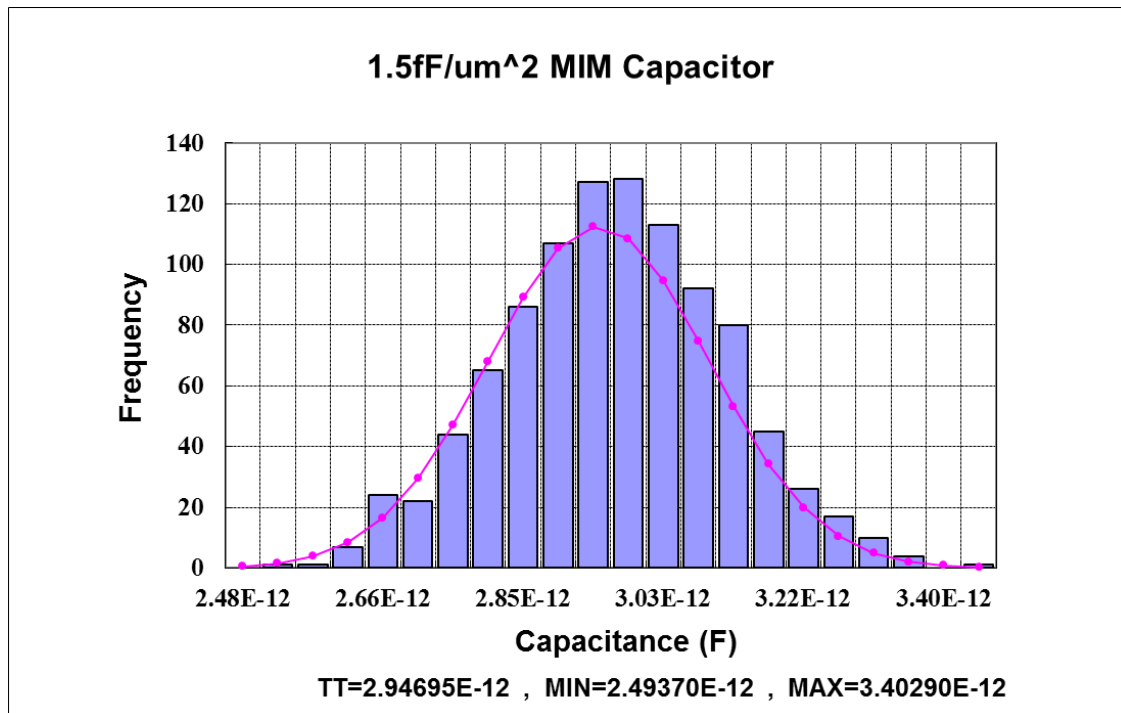


Figure 2-2 Monte Carlo plots of W/L=49.2um/49.2um (before shrink) 1.5fF/um² MIM Capacitor

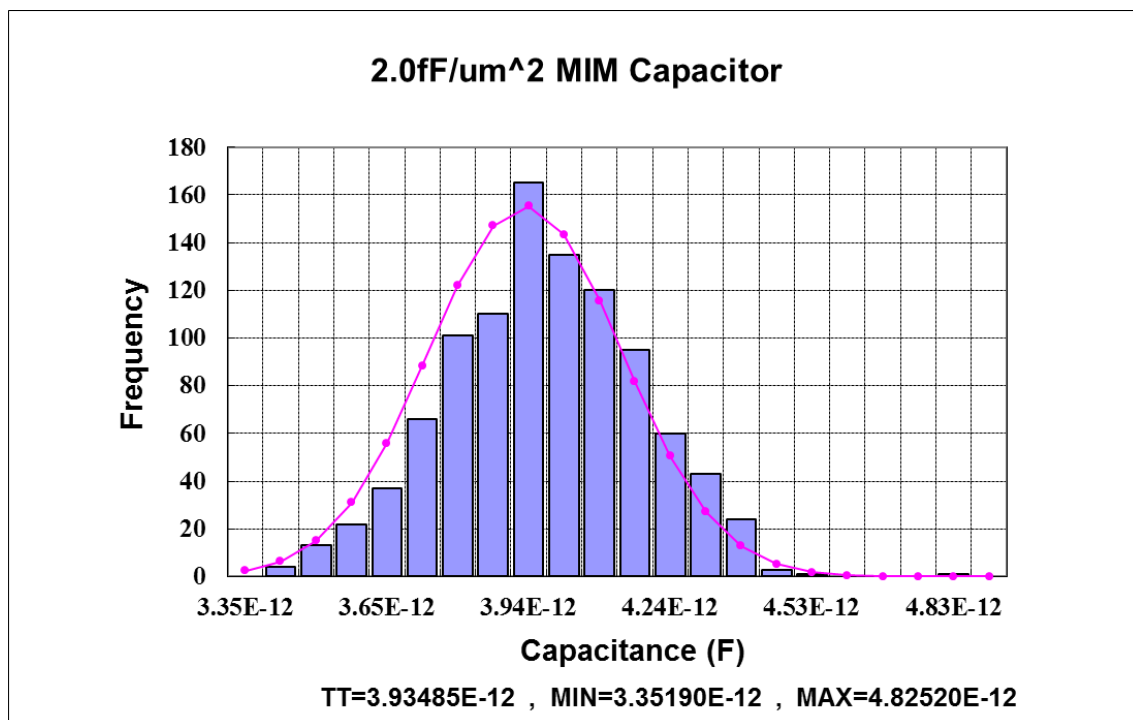


Figure 2-3 Monte Carlo plots of W/L=49.2um/49.2um (before shrink) 2.0fF/um² MIM Capacitor

3 Monte Carlo Simulation Results for Mismatch

3.1 Mis-matching Simulation Result Plots

The following figures show the Monte Carlo mismatch simulation result of Capacitor at 25 °C

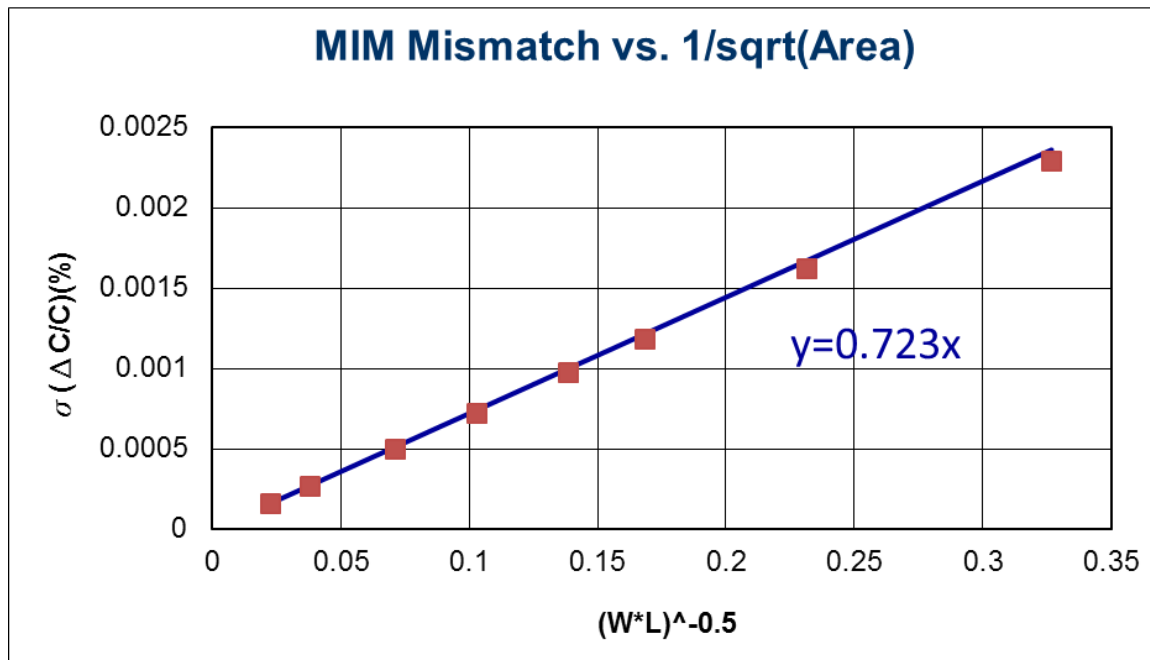


Figure 3-1 Mis-matching simulation vs. 1/sqrt(AREA) of 1.0fF MIM Capacitor

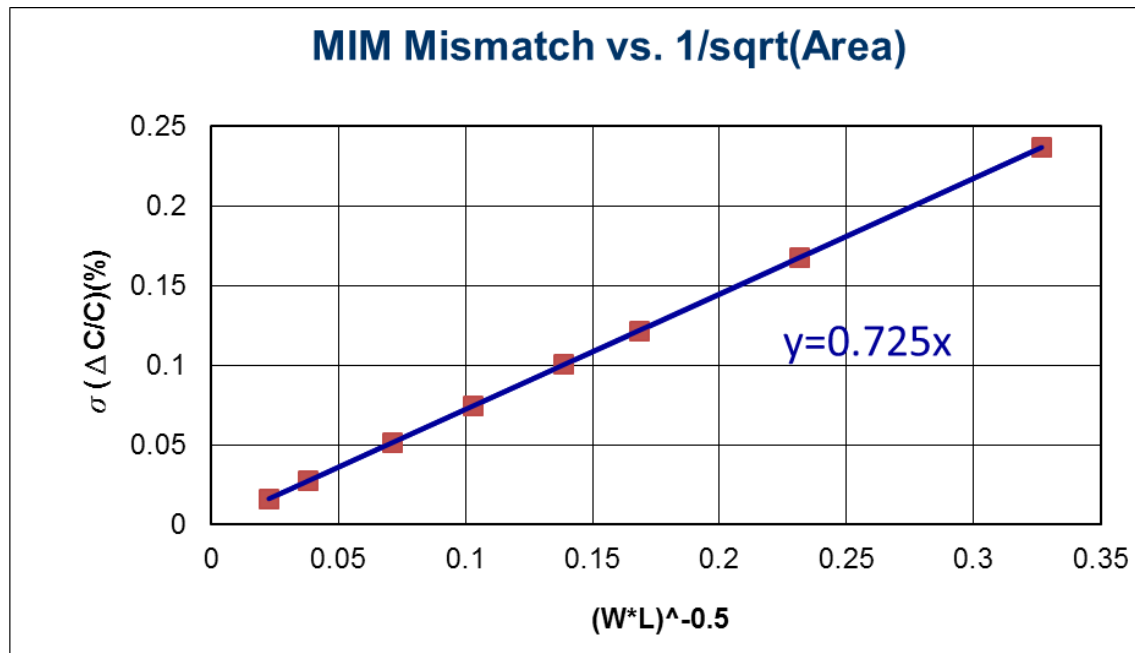


Figure 3-2 Mis-matching simulation vs. 1/sqrt(AREA) of 1.5fF MIM Capacitor

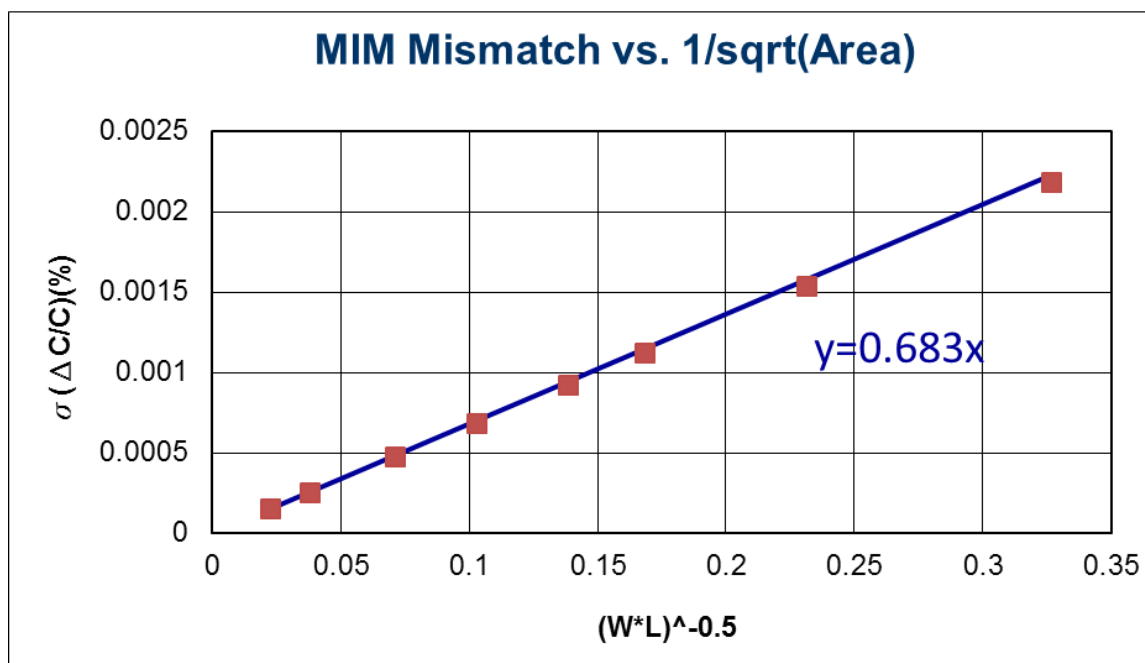


Figure 3-3 Mis-matching simulation vs. 1/sqrt(AREA) of 2.0fF MIM Capacitor

4 Appendix

4.1 HSPICE Warning Messages

Table 4-1 Hspice Warning Message

NO.	Warning Message	Explanation
1.	None	None

4.2 ELDO Warning Messages

Table 4-2 Eldo Warning Message

NO.	Warning Message	Explanation
1.	None	None

4.3 SPECTRE Warning Messages

Table 4-3 Spectre Warning Message

NO.	Warning Message	Explanation
1.	None	None